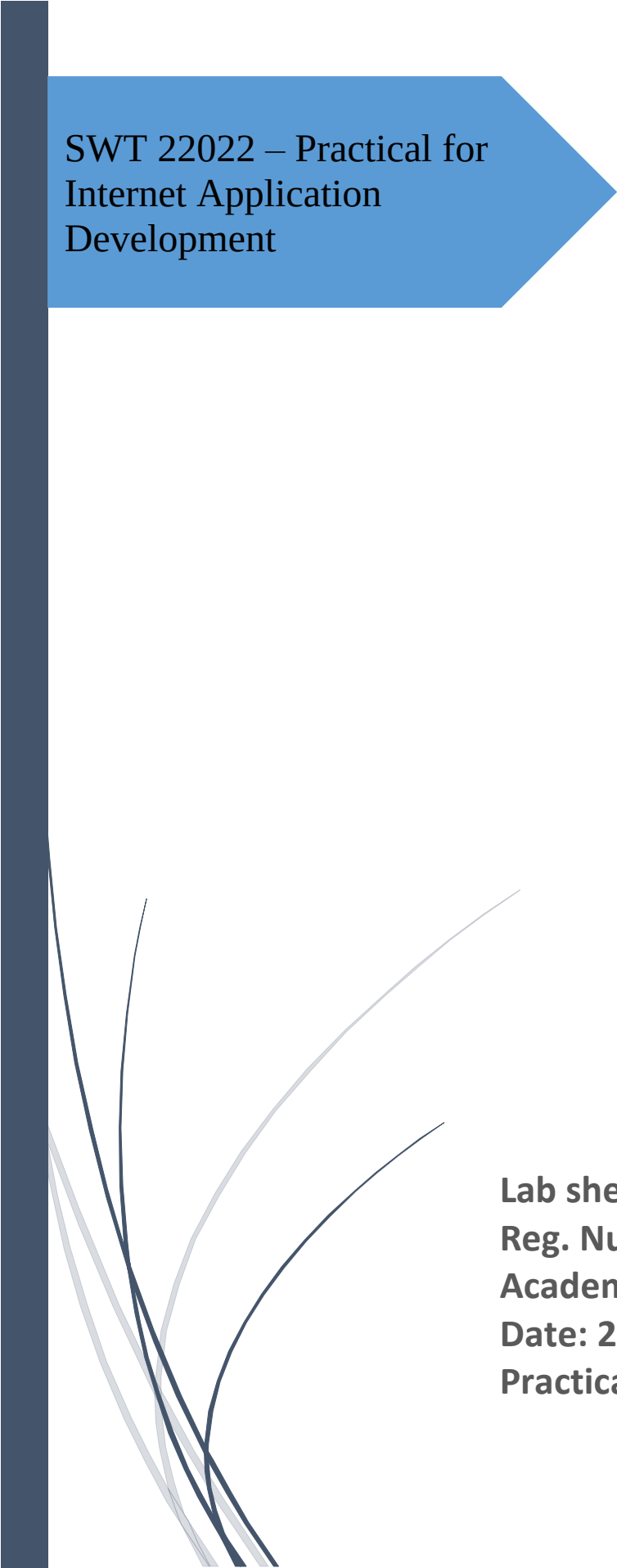




SWT 22022 – Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :05
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Date: 2024.06.24
Practical No :05

Title: PHP Session and Cookies.

Aims:

- Understand and implement PHP sessions and cookies.
- Managing user data and preferences.

Tasks:

- Managing User Sessions.
- User Authentication with sessions.
- Cookies in PHP.
- Working with Cookies in PHP

Activities :

1. Managing User Sessions

```
<?php
    // start a session
    session_start();

    // create session variable
    $_SESSION['username'] = "Olivia123";

    // unset specific session variable data
    unset($_SESSION['username']);

    // unset the all the session variables data
    session_unset();

    // stop the session running
    session_destroy();

?>
```

```

<?php
    session_start();
    $_SESSION['username'] = "Kalhara";
    $_SESSION['index_no'] = "SEU/IS/20/ICT/084";
    $_SESSION['password'] = "#4535fdDED";

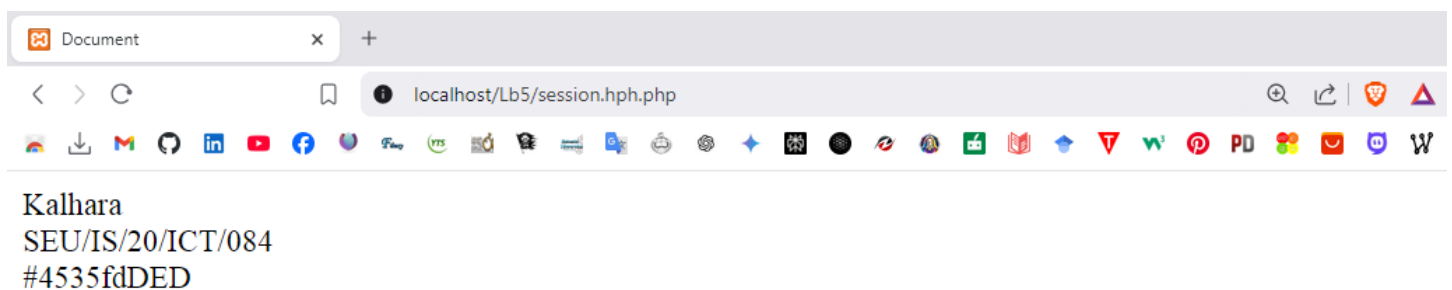
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>

    <?php echo $_SESSION['username']; ?>
    <br>
    <?php echo $_SESSION['index_no']; ?>
    <br>
    <?php echo $_SESSION['password']; ?>

</body>
</html>
<?php
    unset($_SESSION['username']);
    unset($_SESSION['index_no']);
    unset($_SESSION['password']);
    session_unset();
    session_destroy();
?>

```



2. User Authentication with sessions.

I. Create a new PHP file named login.php.

II. Implement login form and php script for store session.

```
<?php
    session_start();
    $username = "Olivia123";
    $password = "Oli123#$2";
    if (isset($_POST['username']) && isset($_POST['password'])) {
        $entered_username = $_POST['username'];
        $entered_password = $_POST['password'];
        if ($entered_username === $username && $entered_password
            ===$password){

            $_SESSION['username'] = $username;
            header("Location: profile.php"); exit;
        }
        else {
            $error_message = "Invalid username or password.";
        }
    }
?>
```

login.php

```
<?php
    session_start();
    $username = "Kalhara";
    $password = "Pasa2001#";

    if (isset($_POST['username']) && isset($_POST['password'])) {
        $entered_username = $_POST['username'];
        $entered_password = $_POST['password'];

        if ($entered_username === $username && $entered_password === $password) {
            $_SESSION['username'] = $username;
            header("Location: profile.php");
            exit;
        } else {
            $error_message = "Invalid username or password.";
        }
    }
?>

<!DOCTYPE html>
<html>
<head>
```

```

    <title>Login Page</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">

</head>
<body>
    <div class="container">
        <div
class="row justify-content-center">
            <div class="col-md-6">
                <h2>Login</h2>

                <?php if (isset($error_message)) { ?>
                    <div class="alert alert-danger" role="alert">
                        <?php echo $error_message; ?>
                    </div>
                <?php } ?>

                <form action="<?php echo $_SERVER['PHP_SELF']; ?>" method="post">
                    <div class="mb-3">

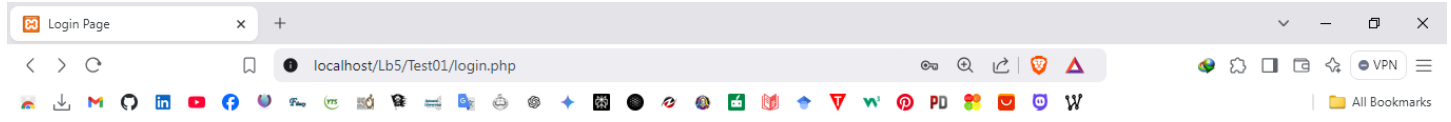
                        <label for="username" class="form-label">Username:</label>
                        <input type="text" class="form-control" id="username"
name="username">
                    </div>

                    <div class="mb-3">
                        <label for="password" class="form-label">Password:</label>
                        <input type="password" class="form-control" id="password"
name="password">
                    </div>

                    <button type="submit" class="btn btn-primary">Login</button>

                </form>
            </div>
        </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></sc
ript>
</body>
</html>

```



Login

Username:

Password:

III. Create a new PHP file named profile.php

IV. Display the name within body tag.

```
<?php
```

```
    session_start();
    if (!isset($_SESSION['username'])) {

        header("Location: login.php");
        exit;
    }
```

```
?>
```

profile.php

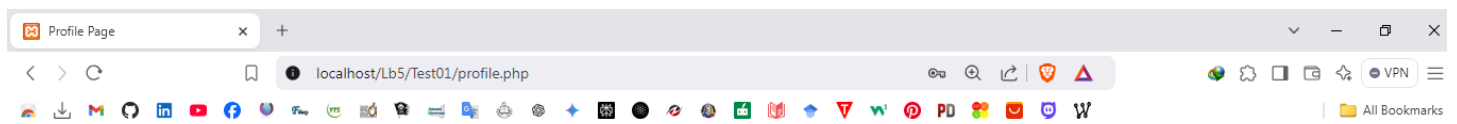
```
<?php
session_start();
if (!isset($_SESSION['username'])) {
    header("Location: login.php");
    exit;
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Profile Page</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">
</head>
```

```

<body>
  <div class="container">
    <div class="row justify-content-center">
      <div class="col-md-6">
        <h2>Welcome to Your Profile, <?php echo
htmlspecialchars($_SESSION['username']); ?>!/h2>
        <p>This is your secure profile page.</p>
        <a href="logout.php" class="btn btn-danger">Logout</a>
      </div>
    </div>
  </div>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>

```



Welcome to Your Profile, Kalhara!

This is your secure profile page.

Logout

V. Create a new PHP file named logout.php

VI. Implement PHP script to logout from the page.

```

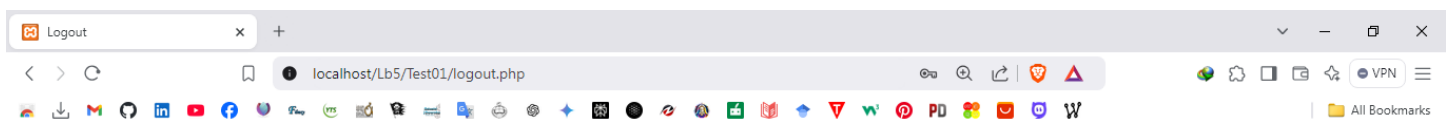
<?php
    session_start();
    session_destroy();
    header("Location: login.php");
    exit;
?>

```

logout.php

```
<?php
session_start();
session_destroy();
?>

<!DOCTYPE html>
<html>
<head>
    <title>Logout</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.m
in.css">
</head>
<body>
    <div class="container">
        <div class="row justify-content-center">
            <div class="col-md-6 text-center">
                <h2>You have been logged out.</h2>
                <p>Thank you for visiting. Click below to log in
again.</p>
                <a href="login.php" class="btn btn-primary">Go to
Login</a>
            </div>
        </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bun
dle.min.js"></script>
</body>
</html>
```



3.Cookies in PHP

```
<?php
    // set cookies
    setcookie("language", "English", time() + (86400 * 5), "/");
    // access cookies
    $language = $_COOKIE['language'];
?>
```

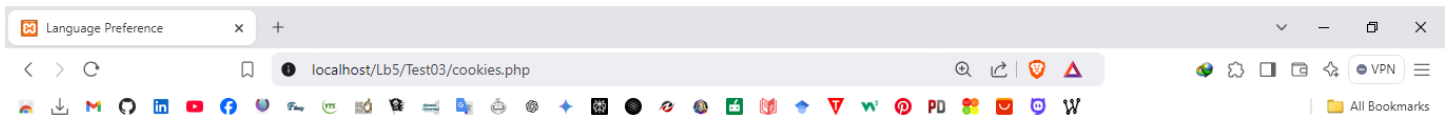
cookies.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Language Preference</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">
</head>
<body>
    <div class="container mt-5">
        <div class="row justify-content-center">
            <div class="col-md-6 text-center">
                <h2>Set Language Preference</h2>

                <?php
                    $cookie_name = "language";
                    $cookie_value = "English";
                    $cookie_expiration = time() + (86400 * 5);
                    setcookie($cookie_name, $cookie_value, $cookie_expiration, "/");

                    if ($_SERVER['REQUEST_METHOD'] === 'POST') {
                        if (isset($_COOKIE[$cookie_name])) {
                            $language = $_COOKIE[$cookie_name];
                            echo "<p>Preferred language: " .
htmlspecialchars($language) . "</p>";
                        } else {
                            echo "<p>Preferred language is not set.</p>";
                        }
                    } else {
                        echo "<form method='POST'><button type='submit' class='btn
btn-primary'>Show Preferred Language</button></form>";
                    }
                ?>
            </div>
        </div>
    </div>
</div>
```

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"
></script>
</body>
</html>
```



4. Working with cookies

I. Create a PHP file named preferred_color.php

II. Implement functionality to remember your preferred color as cookies

```
<?php
```

```
    if (isset($_GET["color_scheme"])) {
        $new_scheme = $_GET["color_scheme"];
        if ($new_scheme === "light" || $new_scheme === "dark") {
            setcookie("color_scheme", $new_scheme, time() + 3600 * 24 *
7);
            header("Location: preferred_color.php");
        }
    }
    $color_scheme = $_COOKIE["color_scheme"];
    if ($color_scheme === "light") {
        $body_style = "background-color: #fff; color: #000;";
    } else {
        $body_style = "background-color: #373A40; color: #fff;";
    }
}
```

```
?>
```

preferred_color.php

```
<?php
if (isset($_GET["color_scheme"])) {
    $new_scheme = $_GET["color_scheme"];
    if ($new_scheme === "light" || $new_scheme === "dark") {
        setcookie("color_scheme", $new_scheme, time() + 3600 * 24 * 7, "/");
        header("Location: preferred_color.php");
        exit;
    }
}
$color_scheme = isset($_COOKIE["color_scheme"]) ? $_COOKIE["color_scheme"] :
"light";
if ($color_scheme === "light") {
    $body_style = "background-color: #fff; color: #000;";
} else {
    $body_style = "background-color: #373A40; color: #fff;";
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Preferred Color Scheme</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">
</head>
<body style="<?php echo $body_style; ?>">
    <div class="container mt-5">
        <div class="row justify-content-center">
            <div class="col-md-6 text-center">
                <h2>Select Your Color Scheme</h2>
                <a href="?color_scheme=light" class="btn btn-light">Light Mode</a>
                <a href="?color_scheme=dark" class="btn btn-dark">Dark Mode</a>
                <hr>
                <p>Your current color scheme is: <strong><?php echo
htmlspecialchars($color_scheme); ?></strong></p>
            </div>
        </div>
    </div>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>
```

Discussion:

- In this lab practical session, I focused on understanding and implementing PHP sessions and cookies, two essential components for managing user data and preferences in web applications. I began by exploring how to manage user sessions effectively, learning how sessions can store user-specific information across multiple pages, enhancing the user experience. The practical aspect involved creating a user authentication system using sessions, allowing users to log in and maintain their authentication status while navigating the site. Additionally, I delved into the role of cookies in PHP, examining how they can store small pieces of data on the client side, such as user preferences and settings. I also practiced working with cookies, implementing functionality to set, retrieve, and delete them. This hands-on experience solidified my understanding of session management and cookie handling in PHP, crucial for developing dynamic and user-friendly web applications.